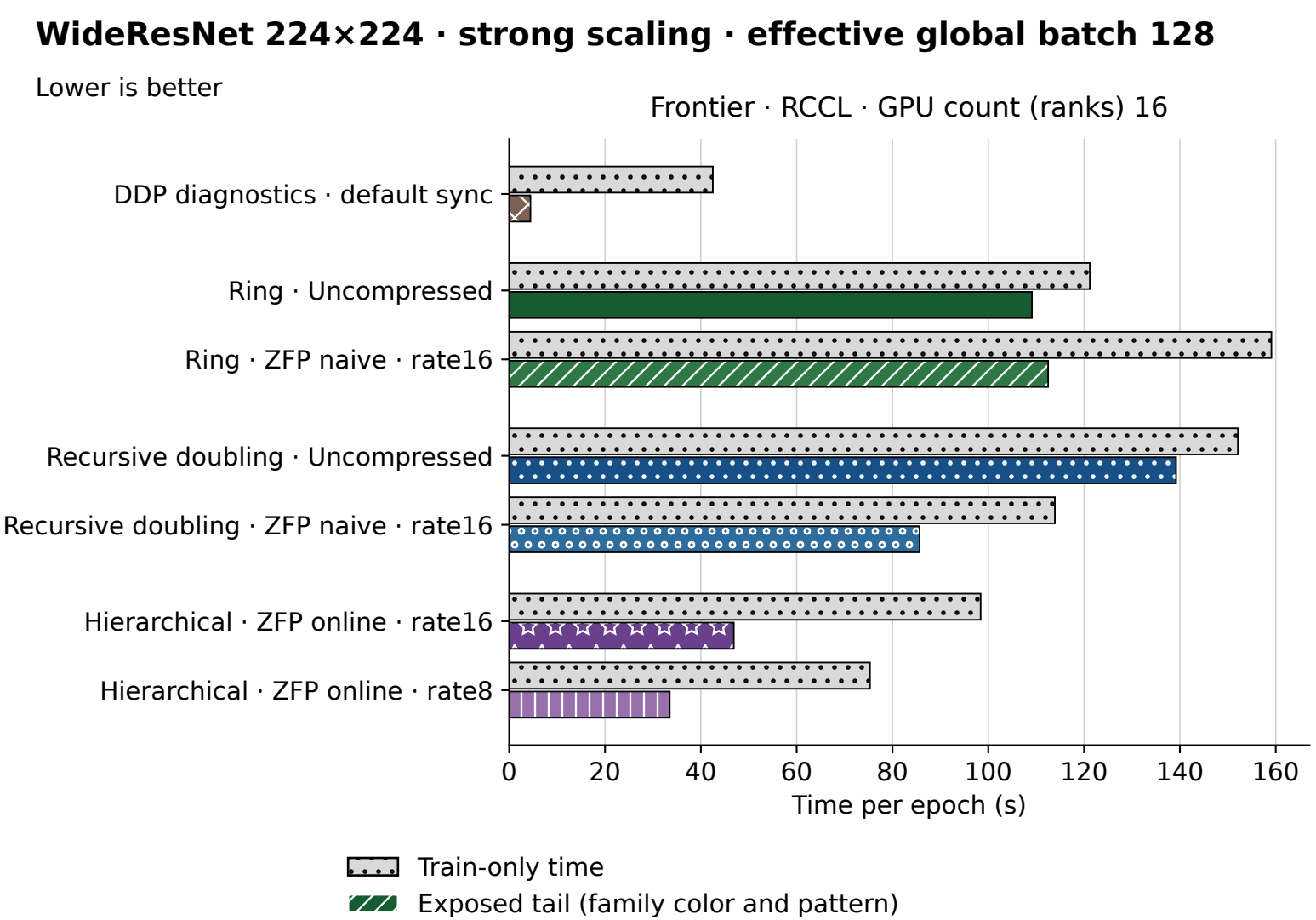

WideResNet 224×224 · strong scaling · effective global batch 128
Lower is better
Frontier · RCCL · GPU count (ranks) 16
DDP diagnostics · default sync
Ring · Uncompressed
Ring · ZFP naive · rate16
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Hierarchical · ZFP online · rate16
Hierarchical · ZFP online · rate8
0
20
40
60
80
100
120
140
160
Time per epoch (s)
Train-only time
Exposed tail (family color and pattern)
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate